9/14 Ch. 2.6

Let $e_1 = \begin{bmatrix} 1 \\ \vdots \\ 0 \end{bmatrix}$ $e_2 = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$ $\cdots$ $e_n = \begin{bmatrix} 0 \\ \vdots \\ 1 \end{bmatrix}$ and $u = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$

$\Rightarrow$ We can express u in terms of : $u = c_1e_1 + c_2e_2 + c_3e_3 \cdots c_ne_n$

$\Rightarrow T(u) = T(c_1e_1 + c_2e_2 \cdots + c_ne_n) = c_1T(e_1) + c_2T(e_2) \cdots + c_nT(e_n) = (T(e_1) + T(e_2) + T(e_3) \cdots) \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ \vdots \\ c_n \end{bmatrix}$

Thus transformation $T(\ )$ of matrix is A, $A = [T(e_1), T(e_2) \cdots T(e_n)]$
A is also called standard matrix.

10/9 Ch. 3.2

Propetis of determinant:

THEOREM 3.2

Let A be an $n \times n$ matrix and c be a nonzero scalar.
(a) If a matrix B is obtained from A by multiplying the elements of a row (column) by c then $|B| = c|A|$.
(b) If a matrix B is obtained from A by interchanging two rows (columns) then $|B| = -|A|$.
(c) If a matrix B is obtained from A by adding a multiple of one row (column) to another row (column), then $|B| = |A|$.

Singular matrix : Determinant of matrix is 0, and it does not have inverse.

Nonsingular matrix : Determinant of matrix is nonzero, and it has inverse.

Let A be a square matrix. A is singular if
(a) all the elements of a row (column) are zero.
(b) two rows (columns) are equal.
(c) two rows (columns) are proportional.

Let A and B be $n \times n$ matrices and c be a nonzero scalar.

(a)	Determinant of a scalar multiple:	$\lvert cA\rvert = c^n\lvert A\rvert$
(b)	Determinant of a product:	$\lvert AB\rvert = \lvert A\rvert\lvert B\rvert$
(c)	Determinant of a transpose:	$\lvert A^t\rvert = \lvert A\rvert$
(d)	Determinant of an inverse:	$\lvert A^{-1}\rvert = \frac{1}{\lvert A\rvert}$ (Assuming A^{-1} exists)